

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: COMPUTER VISION with OpenCV

COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)*

Assessment Tool Weightage: 20%

Scenario-Based Questions

1. Low Contrast Image

(a) Interpretation

A grayscale image appears dull because there is very little difference between the intensity values of the object and the background. This makes it difficult to distinguish image features.

(b) Key Issues

- Poor visibility of objects.
- Narrow intensity range.
- Reduced image contrast.
- Difficulty in feature extraction and segmentation.

(c) Enhancement Technique

Histogram Equalization

Histogram Equalization redistributes pixel intensities over the entire available intensity range, thereby increasing image contrast.

(d) Justification

- Enhances global contrast effectively.
- Improves visibility of hidden details.
- Makes objects easier to identify.
- Simple and computationally efficient.

(e) Conclusion

Histogram Equalization is the most suitable enhancement method because it improves contrast, reveals hidden features, and increases image quality for further processing.

2. Uneven Illumination

(a) Interpretation

The image contains bright and dark regions due to non-uniform lighting during image acquisition, resulting in inconsistent brightness.

(b) Key Issues

- Non-uniform illumination.
- Bright and dark patches.
- Reduced segmentation accuracy.
- Uneven intensity distribution.

(c) Enhancement Technique

Adaptive Histogram Equalization (CLAHE)

CLAHE enhances local contrast while preventing excessive amplification of noise.

(d) Justification

- Corrects local brightness variations.
- Preserves image details.
- Produces balanced illumination.
- Suitable for images with varying lighting conditions.

(e) Conclusion

CLAHE effectively normalizes illumination and improves visibility across the entire image.

3. Salt-and-Pepper Noise

(a) Interpretation

The scanned document contains random black and white pixels caused by transmission or sensor errors, reducing readability.

(b) Key Issues

- Random impulse noise.
- Loss of image clarity.
- Reduced readability.
- Interference in text recognition.

(c) Filtering Technique

Median Filter

The median filter replaces each pixel with the median value of its neighboring pixels.

(d) Justification

- Removes salt-and-pepper noise efficiently.
- Preserves edges better than mean filtering.
- Does not blur image details significantly.

(e) Conclusion

The Median Filter is the best choice because it removes impulse noise while maintaining important image structures.

4. Object Boundary Detection

(a) Interpretation

The objective is to accurately detect tissue boundaries in a medical image for diagnosis and analysis.

(b) Key Challenges

- Weak edges.
- Image noise.
- Similar intensity values between tissues.
- Blurred boundaries.

(c) Edge Detection Method

Canny Edge Detection

Steps:

1. Noise reduction using Gaussian filter.
2. Gradient computation.
3. Non-maximum suppression.
4. Double thresholding.
5. Edge tracking by hysteresis.

(d) Justification

- Detects edges accurately.
- Reduces false edge detection.
- Produces thin and continuous edges.
- Performs well even in noisy images.

(e) Conclusion

Canny Edge Detection provides precise and reliable boundary detection, making it ideal for medical image analysis.

5. Combined Requirement

(a) Interpretation

An industrial inspection system requires removing noise while preserving important edges to accurately detect defects.

(b) Key Issues

- Presence of image noise.
- Loss of edge information.
- Reduced defect detection accuracy.

(c) Sequence of Techniques

1. Apply **Median Filter** (or Bilateral Filter) for noise removal.
2. Apply **Canny Edge Detection** to detect edges.
3. Perform segmentation if required for defect identification.

(d) Justification

- Noise removal improves image quality.
- Edge preservation ensures important defect boundaries remain visible.
- Sequential processing improves overall inspection accuracy.

(e) Conclusion

Applying noise filtering followed by edge detection provides high-quality images suitable for reliable industrial defect detection.

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:**RUBRICS FOR CALCULATION:**

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues

Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes wellreasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

		presented			
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